

Online Appendix

Mapping O*NET Tasks to ISCO Occupations using Text Similarity

Majid Einian

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A Parameter selection

Parameter values are selected through a multi-phase unsupervised search over blend weights that determine the share of each text source in the final query and target embeddings. The sweep is conducted on O*NET 29.2, which was the most recent release of the SOC 2018 taxonomy series at the time of parameter selection. Because the five blend weights control only how pre-computed component embeddings are combined—not any feature specific to a particular O*NET release—the identified values transfer unchanged to all 63 O*NET releases; this transferability is corroborated by the stable validation results across versions reported in Appendix C. The selection procedure is described here before the validation results (Appendices B–D), as the parameter selection is performed entirely on unsupervised criteria; validation results are examined afterwards as a consistency check only.

Search procedure

An adaptive iterative sweep is conducted over the five blend weights: w_{soc} , w_{dwa} , w_{isco} , $w_{\text{isco-task}}$, $w_{\text{occ}} \in [0, 1]$. All variants share pre-computed embeddings for the SOC occupation titles, ISCO-08 task descriptions, and ESCO occupation text and skill inventory; the variable cost is only the fast blend and FAISS retrieval step. One feasibility constraint is enforced: $w_{\text{soc}} + w_{\text{dwa}} \leq 0.90$, keeping at least 10% weight on the raw task text in the query vector.

Round 1 (initial grid). Five uniformly-spaced values are placed across each parameter’s full range, yielding a $5^5 = 3,125$ candidate grid reduced to 1,250 feasible configurations after the query-weight constraint.

Rounds 2–6 (adaptive zoom). After each round all five parameters are re-gridded simultaneously using a trust-region strategy based on the current local step h and the current best value b for each parameter:

- *Local grid boundary, not global bound:* expand by one full step, targeting $[b - 2h, b - h, b, b + h, b + 2h]$ (clipped to $[0, 1]$).
- *True global bound:* one-sided refinement with a halved step $h' = \max(h/2, h_{\min})$, where the step-balance floor $h_{\min} = \max_p(h_p)/4$ prevents any one dimension from becoming much finer than the others.
- *Interior:* symmetric refinement with halved step h' , placing five points at $[b - 2h', b - h', b, b + h', b + 2h']$.

The sweep will not converge if any parameter is at a grid boundary or if any parameter has not yet been refined below its initial step size, ensuring at least one interior-zoom pass for each dimension. If clipping at $[0, 1]$ reduces a parameter’s grid to fewer than five distinct points, additional points are placed at successive multiples of the current step around the best value until five distinct points lie within bounds. All five parameters are re-gridded simultaneously after each round; no coordinate-wise or sequential updating is used. The sweep is fully deterministic: component embeddings are cached on disk and reloaded identically across runs, and the feasible configuration set is enumerated in a fixed iteration order. When multiple configurations share the maximum sweep score within a round, the configuration encountered first in that order is used as the reference for the next zoom; in the six rounds run here, ties of this kind did not arise. The sweep terminates when the improvement in the best sweep score of the current round over the global best from all preceding rounds falls below 0.0005, *and* every parameter’s current best value is strictly interior to its grid, *and* every parameter’s current

step is strictly below its Round 1 value of 0.25. The threshold 0.0005 corresponds approximately to a 0.15 percentage-point change in coverage, which is below the noise floor of a single ISCO coverage step.

The six rounds yield 16,000 candidate specifications in total. The per-round configuration counts decrease as the grid concentrates around the optimum: 1,250 (Round 1), 3,000 (Round 2), 2,375 (Round 3), 3,125 (Round 4), 3,125 (Round 5), and 3,125 (Round 6).

Sweep score

Each candidate is scored by

$$\text{sweep_score} = \frac{3 \cdot \text{cov} + 2 \cdot \text{sim} - 2 \cdot \text{overload} - 2 \cdot \text{gini}}{9},$$

where cov is the ISCO-08 coverage share (share of 4-digit ISCO-08 unit groups receiving at least one task assignment), sim is the mean cosine similarity of retained task-ISCO links, overload is the share of tasks assigned to overloaded ISCO groups, and gini is the Gini coefficient of the task-count distribution across ISCO groups (Table A1). The formula is applied identically in every round without re-normalisation, so scores are directly comparable across rounds.

An ISCO-08 unit group is classified as overloaded if its task count exceeds $T = \max(200, Q_{0.95})$, where $Q_{0.95}$ is the 95th percentile of the per-group task-count distribution for the configuration being evaluated. The absolute floor of 200 ensures the threshold is not trivially tight in configurations with high concentration. The overload penalty term in the sweep score penalises parameter configurations that produce excessive task concentration, steering selection away from weight combinations in which broad ISCO groups absorb semantically unrelated tasks. Among the 16,000 candidate configurations, the overload share ranges from 1.2% to 31.0% (std 2.9 pp), confirming that this criterion actively discriminates between parameter configurations.

Table A1: Sweep score components.

Metric	Weight	Direction
ISCO-08 coverage share	3	maximize
Mean cosine similarity retained	2	maximize
Share of tasks in overloaded ISCO group	2	minimize
Gini coefficient of task count per ISCO group	2	minimize

The 3-2-2-2 weight scheme reflects a priority ordering fixed *before* running the sweep and was not itself optimised. Coverage receives the highest weight because any uncovered ISCO-08 unit group represents a hard gap in the crosswalk: without at least one task assignment, no task-based exposure estimate can be produced for that occupation group. Mean cosine similarity receives weight 2 as a direct quality signal for retained links. Overload and Gini each receive penalty weight 2 as complementary concentration guards. The overload term detects configurations where any single group becomes a dominant absorber (task counts above the data-adaptive threshold T); the Gini term penalises uneven distribution across all groups. Together they steer the sweep away from configurations with pathological concentration, which arise at extreme parameter values and are characterised by lower mean similarity and up to 9–13 overloaded groups with task counts exceeding 700. The denominator 9 equals the sum of absolute weights ($3 + 2 + 2 + 2$); the score therefore ranges from $-4/9 \approx -0.44$ (no coverage, no similarity, full overload, maximal Gini) to $5/9 \approx 0.56$ (full coverage, maximum similarity, zero overload, zero Gini). All findings about optimal weight values in this section are conditional on this objective function. Alternative weighting schemes were not tested; placing greater emphasis on similarity, for instance, would shift the identified optimum toward configurations with fewer but higher-quality links.

Selected configuration

The configuration with the highest sweep score across all 16,000 candidates is selected as the production specification: $w_{\text{soc}} = 0.375$, $w_{\text{dwa}} = 0.266$, $w_{\text{isco}} = 0.375$, $w_{\text{isco-task}} = 0.734$, $w_{\text{occ}} = 0.156$ (sweep score 0.3907, labelled **r6_s375_d266_i38_t73_o16**; coverage 99.8%, mean similarity 0.748, Gini 0.457, overloaded share 2.9%). The low overloaded share reflects structural SOC–ISCO-08 taxonomy asymmetry (see Appendix E) rather than a quality problem requiring pipeline correction: the two groups exceeding the concentration threshold (ISCO 2310, ISCO 1345) each draw over 75% of their tasks from a single SOC major group, confirming a structural rather than catch-all origin. Table A2 shows the top configurations by sweep score. The top candidates all achieve ISCO-08 coverage of 99.8–100% and span a narrow range of sweep scores, confirming that the crosswalk is insensitive to small perturbations in these weights under the sweep-score objective. This robustness is with respect to the unsupervised objective function and does not directly imply robustness of external validation performance.

Table A2: Top configurations by sweep score (selection score is a normalised composite used only for ranking within the sweep; raw sweep scores are comparable across rows).

Rank	Changed parameter	Value	Selection score	S3 coverage	S3 mean similarity	S3 Gini
1	baseline	baseline	0.832	1.000	0.718	0.446
2	multiple	$w_{\text{dwa}}=0.0$; $w_{\text{isco_task}}=0.875$	0.832	1.000	0.700	0.434
3	w_{isco}	0.4375	0.831	1.000	0.718	0.448
4	multiple	$w_{\text{dwa}}=0.1875$; $w_{\text{isco}}=0.4688$; $w_{\text{isco_task}}=0.7812$	0.831	1.000	0.725	0.452
5	multiple	$w_{\text{dwa}}=0.0$; $w_{\text{isco_task}}=1.0$	0.830	1.000	0.697	0.430
6	multiple	$w_{\text{dwa}}=0.1875$; $w_{\text{isco}}=0.4062$; $w_{\text{isco_task}}=0.7812$	0.829	1.000	0.726	0.453
7	multiple	$w_{\text{soc_title}}=0.125$; $w_{\text{isco_task}}=1.0$	0.828	1.000	0.695	0.436
8	multiple	$w_{\text{occ}}=0.0$; $w_{\text{isco_task}}=0.75$	0.828	1.000	0.717	0.444
9	multiple	$w_{\text{dwa}}=0.0$; $w_{\text{occ}}=0.0$; $w_{\text{isco_task}}=0.875$	0.827	1.000	0.698	0.431
10	multiple	$w_{\text{occ}}=0.0$; $w_{\text{isco_task}}=0.875$	0.827	1.000	0.716	0.441
11	$w_{\text{isco_task}}$	0.875	0.827	1.000	0.718	0.443
12	$w_{\text{isco_task}}$	0.875	0.827	1.000	0.718	0.443

The chosen configuration can be read directly from its weights. On the *query side*, the blend is constructed in two sequential steps. First, the raw task embedding and the DWA label embedding are combined: $\text{core} = (1 - w_{\text{dwa}}) \cdot \text{task} + w_{\text{dwa}} \cdot \text{DWA}$. Second, the SOC title embedding is blended in: $\text{query} = (1 - w_{\text{soc}}) \cdot \text{core} + w_{\text{soc}} \cdot \text{title}$. In the implementation, component vectors are L2-normalised before blending and the query vector is L2-normalised after each blend, so the weights should be interpreted as nominal blend weights rather than exact post-normalisation vector shares. Because the dials are applied sequentially, the effective contributions of each source are: SOC title 37.5% ($= w_{\text{soc}}$), raw task text 45.9% ($= (1 - w_{\text{soc}})(1 - w_{\text{dwa}})$), and DWA labels 16.6% ($= (1 - w_{\text{soc}}) \cdot w_{\text{dwa}}$). All three sources play a meaningful role, with the raw task text providing the largest share of task-specific content, the SOC title anchoring occupational context, and DWA labels supplying a complementary functional signal.

On the *target side*, $w_{\text{isco}} = 0.375$ allocates 37.5% of the target representation to ISCO-08 text and 62.5% to the ESCO component. Within the ISCO component, $w_{\text{isco-task}} = 0.734$ gives 73.4% weight to ISCO task-item descriptions versus 26.6% to the generic group description. Within the ESCO component, $w_{\text{occ}} = 0.156$ allocates only 15.6% to occupation descriptions and 84.4% to the ESCO skill inventory. In nominal target-side weights, the ESCO skill inventory is the single largest contributor to the target embedding (52.7%), followed by ISCO task items (27.5%), ISCO information text (10.0%), and ESCO occupation text (9.8%). The target vector is L2-normalised before retrieval. Within the selected optimization framework, the ESCO skill inventory receives the largest nominal weight among all target-side components, suggesting that skill-level vocabulary provides a useful semantic anchor for matching O*NET tasks to ISCO-08 unit groups under the chosen objective.

Figures A1 and A2 show the sweep score surface for the target-side and query-side parameters respectively. Each cell reports the maximum sweep score across all configurations sharing that pair of parameter values (marginalising over the remaining parameters). The selected configuration is marked with a red star.

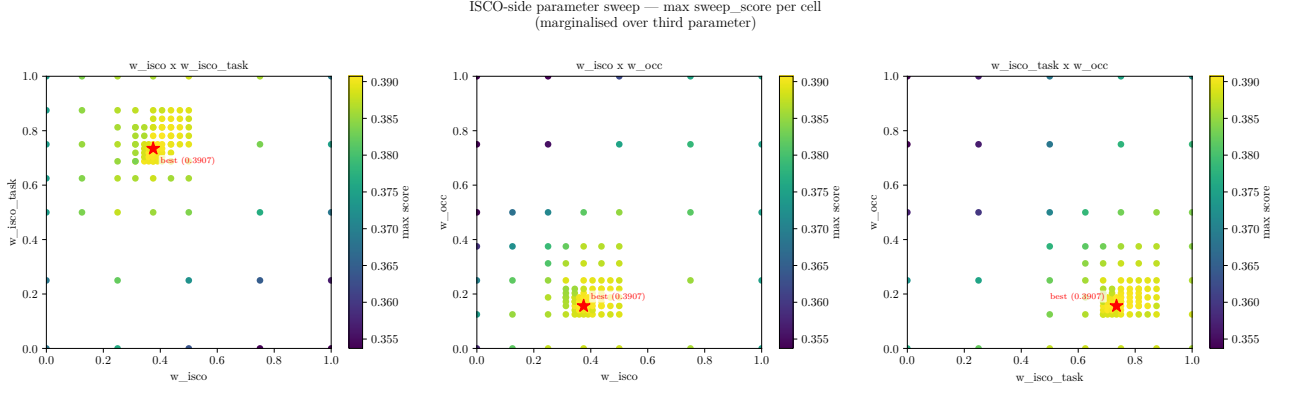


Figure A1: Target-side parameter sweep: maximum sweep score per cell for pairwise combinations of w_{isco} , $w_{\text{isco-task}}$, and w_{occ} (third parameter marginalised to its best-case value). Each cell therefore reports an upper bound on performance for that pair, not an average across all values of the remaining parameter. Red star marks the selected configuration.

Query-side (task) parameter sweep — max sweep_score per cell

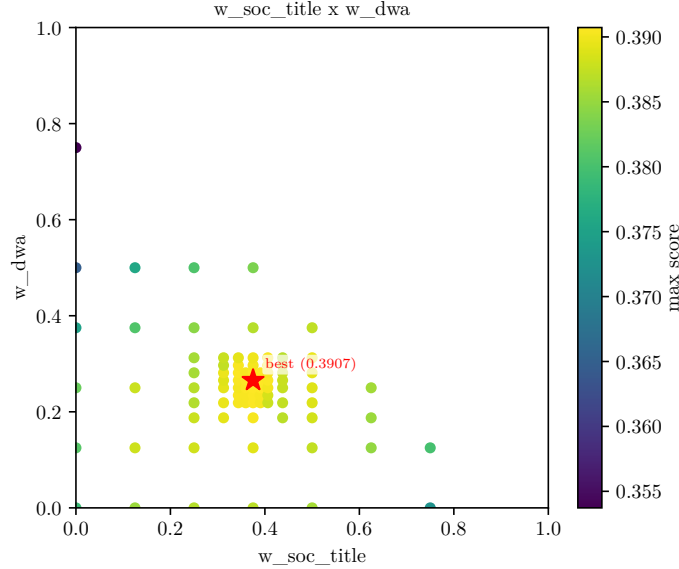


Figure A2: Query-side parameter sweep: maximum sweep score per cell for w_{soc} and w_{dwa} (target-side parameters marginalised to their best-case values; each cell reports an upper bound for that pair). Red star marks the selected configuration.

Findings

The sweep reveals that DWA labels provide a meaningful positive contribution at $w_{\text{dwa}} \approx 0.27$, which translates to an effective contribution of $\approx 16.6\%$ of the query vector: neither zero nor dominant DWA weight is optimal, indicating that functional task descriptors complement but do not replace the raw task text and SOC title anchors.

On the target side, the ESCO skill inventory dominates the optimal representation (effective weight $\approx 52.8\%$ of the full target vector), while ISCO task descriptions contribute $\approx 27.5\%$. Within the chosen objective function, skill-level vocabulary appears to bridge the gap between O*NET task language and ISCO-08 unit groups more effectively than the official ISCO-08 task descriptions alone. The relatively low weight on ISCO-08 text ($w_{\text{isco}} = 0.375$) is consistent with the sparseness of the ISCO-08 task lists for some unit groups, which limits their discriminative power when used in isolation.

Each task is assigned to a single ISCO-08 unit group. Allowing multiple assignments increases coverage but reduces interpretability and complicates aggregation of task-based measures. Configurations

with high sweep scores cluster in a narrow region of the parameter space (see Table A2), indicating that the selected configuration is not a knife-edge optimum with respect to the unsupervised objective.

B Validation framework

Validation compares occupation-level assignments implied by the task-based mapping to institutional SOC–ISCO concordances constructed by statistical agencies and international organisations. Task-level assignments are first aggregated to the SOC occupation level. Each SOC occupation is assigned the ISCO-08 unit group receiving the largest number of task matches. Agreement is then evaluated relative to institutional SOC–ISCO mappings at both the four-digit unit-group level and the one-digit major-group level.

Multiple institutional concordances are used in order to avoid dependence on any single mapping source. Agreement is evaluated relative to both strict and lenient benchmarks, defined formally as follows. Let $\mathcal{C}(s)$ denote the set of institutional crosswalks available for SOC occupation s , and let $I_c(s)$ denote the set of ISCO-08 unit groups that crosswalk c assigns to occupation s . The *strict benchmark* accepts the pipeline’s assignment $\hat{l}(s)$ only if it falls in every available crosswalk’s set:

$$I^{\text{strict}}(s) = \bigcap_{c \in \mathcal{C}(s)} I_c(s).$$

The *lenient benchmark* accepts $\hat{l}(s)$ if it falls in at least one:

$$I^{\text{lenient}}(s) = \bigcup_{c \in \mathcal{C}(s)} I_c(s).$$

The four-digit in-set rate is the share of SOC occupations s for which $\hat{l}(s) \in I^\bullet(s)$. Results are similar across benchmark definitions, indicating that agreement rates are not driven by any single concordance.

Institutional crosswalks are used only for validation and do not influence construction of the task-level mapping.

C Full validation results

The pipeline was applied to all O*NET releases from v4.0 (2001) through v30.3 (2025), covering 63 releases in total across five SOC taxonomy generations (pre-2006, SOC 2006, SOC 2009, SOC 2010, SOC 2018). Table C1 reports chain crosswalk validation results for alternative benchmark definitions across three representative releases per major SOC taxonomy era: O*NET 25.1, 29.2, and 30.3 for the SOC 2018 generation (first, parameter-selection, and latest releases); and O*NET 15.1, 20.0, and 25.0 for the SOC 2010 generation (first, mid-era, and last releases). These two eras are shown because the institutional crosswalks available for validation differ between them: the ESCO–ONET–SOC chain crosswalks cover SOC 2018 (O*NET 25.1 onward), while the ESCO–O*NET and BLS–ISCO crosswalks cover SOC 2010. The main text reports results for O*NET 29.2 using the lenient benchmark, which balances measurement error across institutional concordances.

Table C1: Chain crosswalk validation: match rates across scenarios and representative O*NET releases.

Scenario	Cov. (%)	Exact (%)	Sub- major (%)	Major group (%)
<i>SOC 2018 — O*NET 25.1 (first SOC 2018 release)</i>				
A1 – ESCO-SOC18	99.9	67.7	81.0	88.3
A2 – SOC18-ESCO	98.7	59.7	73.9	81.5
A3 – Strict intersection	98.5	59.6	73.8	81.5

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Table C1 – continued

Scenario	Cov. (%)	Exact (%)	Sub- major (%)	Major group (%)
A4 – Lenient union	99.9	67.9	81.2	88.4
<i>SOC 2018 — O*NET 29.2 (release used for parameter selection)</i>				
A1 – ESCO-SOC18	99.9	67.8	80.9	88.3
A2 – SOC18-ESCO	98.7	59.7	74.0	81.5
A3 – Strict intersection	98.5	59.6	73.8	81.5
A4 – Lenient union	99.9	68.0	81.2	88.4
<i>SOC 2018 — O*NET 30.3 (latest release)</i>				
A1 – ESCO-SOC18	99.9	67.7	80.9	88.2
A2 – SOC18-ESCO	98.7	59.7	73.9	81.5
A3 – Strict intersection	98.5	59.6	73.8	81.4
A4 – Lenient union	99.9	67.9	81.2	88.4
<i>SOC 2010 — O*NET 15.1 (first SOC 2010 release)</i>				
B1 – ESCO-ONET-MHV ^a	79.1	48.7	64.1	73.5
B2 – SOC10-ISCO-BLS	99.9	43.2	58.0	67.2
B3 – Lenient union (MHV $\cup$ BLS)	100.0	48.1	64.2	72.9
B4 – Strict intersection (MHV $\cap$ BLS)	70.0	47.8	61.3	69.7
<i>SOC 2010 — O*NET 20.0 (mid-era)</i>				
B1 – ESCO-ONET-MHV ^a	79.3	48.6	64.3	73.6
B2 – SOC10-ISCO-BLS	99.9	42.9	57.9	67.2
B3 – Lenient union (MHV $\cup$ BLS)	100.0	48.0	64.3	73.0
B4 – Strict intersection (MHV $\cap$ BLS)	70.2	47.5	61.2	69.7
<i>SOC 2010 — O*NET 25.0 (last SOC 2010 release)</i>				
B1 – ESCO-ONET-MHV ^a	79.1	50.2	65.9	74.8
B2 – SOC10-ISCO-BLS	99.9	44.3	59.8	68.7
B3 – Lenient union (MHV $\cup$ BLS)	100.0	49.5	66.1	74.3
B4 – Strict intersection (MHV $\cap$ BLS)	70.0	49.3	63.1	71.2
^a Less independent: derived via the same semantic similarity approach as the pipeline.				

Table C2 shifts from task-level to occupation-level validation. Whereas Table C1 evaluates each task’s ISCO assignment individually, Table C2 collapses the pipeline output to one ISCO per SOC occupation (the mode across tasks in that SOC) and then compares this implied SOC→ISCO mapping against institutional crosswalks. This complements Table C1 by checking whether the pipeline’s aggregate occupational mapping is coherent with what expert-built crosswalks say, independently of the task-level scatter.

The table is restricted to the same six representative releases as Table C1. Two metrics are reported. *Pair precision* is the share of (SOC, ISCO) pairs implied by the pipeline that appear in the institutional crosswalk; *in-set rate* (4-digit, 2-digit, 1-digit) is the share of SOC occupations for which the pipeline’s top-1 ISCO falls anywhere in the crosswalk’s acceptable set at that level of the hierarchy.

Pair precision is structurally low: the pipeline assigns exactly one ISCO per SOC occupation, while institutional crosswalks typically list several acceptable ISCO codes per SOC. An occupation where the pipeline picks the correct ISCO from among five reference options contributes $1/5 = 0.20$ to pair precision but 1.00 to the in-set rate. In-set rate is therefore the appropriate primary accuracy measure; pair recall and F1 are included for completeness.

SOC 2010 versions (O*NET 15.1–25.0) are evaluated against the ESCO–O*NET and BLS–ISCO crosswalks; SOC 2018 versions (O*NET 25.1–30.3) are evaluated against the ESCO–ONET–SOC crosswalks. These are different instruments with different many-to-many cardinalities, so pair precision and in-set rates should not be compared directly across the two SOC eras.

Table C2: Occupation-level agreement with institutional crosswalks (same six releases as Table C1).

Dataset	Institutional crosswalk	Pair precision	Pair recall	Pair F1	In-set (4-digit)	In-set (2-digit)	In-set (1-digit)	Shared SOC n
O*NET 15.1	ESCO-ONET (MHV)	0.129	0.711	0.219	0.708	0.785	0.825	576
O*NET 15.1	BLS ISCO-SOC	0.147	0.713	0.243	0.650	0.721	0.773	774
O*NET 20.0	ESCO-ONET (MHV)	0.127	0.710	0.216	0.708	0.785	0.826	576
O*NET 20.0	BLS ISCO-SOC	0.145	0.715	0.241	0.652	0.722	0.775	774
O*NET 25.0	ESCO-ONET (MHV)	0.131	0.702	0.221	0.733	0.811	0.852	576
O*NET 25.0	BLS ISCO-SOC	0.150	0.713	0.249	0.678	0.756	0.804	774
O*NET 25.1	ESCO-ONET-SOC (SOC18)	0.340	0.468	0.394	0.868	0.910	0.944	797
O*NET 25.1	ONET-SOC-ESCO (SOC18)	0.246	0.615	0.351	0.811	0.861	0.897	784
O*NET 29.2	ESCO-ONET-SOC (SOC18)	0.343	0.464	0.394	0.868	0.911	0.944	797
O*NET 29.2	ONET-SOC-ESCO (SOC18)	0.248	0.608	0.352	0.812	0.862	0.898	784
O*NET 30.3	ESCO-ONET-SOC (SOC18)	0.343	0.464	0.394	0.867	0.910	0.942	797
O*NET 30.3	ONET-SOC-ESCO (SOC18)	0.247	0.608	0.352	0.811	0.861	0.897	784

Table C3 reports internal consistency between pairs of institutional crosswalks for the same SOC version. High within-version agreement between independent reference sources indicates that the institutional benchmarks themselves are coherent.

Table C3: Internal consistency between pairs of institutional crosswalks.

SOC version	Reference A	Reference B	Pair precision	Pair recall	Pair F1	In-set (4-digit)	In-set (2-digit)	In-set (1-digit)
SOC 2010	ESCO-ONET (MHV)	BLS ISCO-SOC	0.613	0.543	0.576	0.810	0.856	0.895
SOC 2018	ESCO-ONET-SOC (SOC18)	ONET-SOC-ESCO (SOC18)	0.544	0.988	0.701	0.598	0.690	0.738

Table C4 lists the top mismatches between the pipeline and institutional crosswalks for one representative release per SOC era: O*NET 25.0 against the BLS ISCO-SOC crosswalk, and O*NET 29.2 against the ESCO-ONET-SOC crosswalk. These examples should be read as suggestive rather than definitive: disagreement can originate from noise in the reference crosswalk rather than from the pipeline. A case in point is *Proofreaders and Copy Markers* (SOC 43-9081, O*NET 29.2): the pipeline assigns ISCO 4413 (*Coding, Proofreading and Related Clerks*), which directly matches the occupational tasks; the crosswalk maps this SOC to ISCO 2642 (*Journalists*), a pairing that is difficult to justify on task-content grounds. Similarly, *Survey Researchers* (SOC 19-3022) are assigned by the pipeline to ISCO 4227 (*Survey and Market Research Interviewers*), while the BLS crosswalk places them in ISCO 2120 (*Mathematicians, Actuaries and Statisticians*). In cases such as these the pipeline assignment is arguably more accurate than the reference, so the in-set rate provides a lower bound on true classification quality.

Table C4: Illustrative mismatches between the pipeline and institutional crosswalks (four examples per release; each row shows the top-1 ISCO from both available reference crosswalks).

SOC occupation	Pipeline ISCO	Ref. 1 ISCO	Ref. 2 ISCO
<i>O*NET 25.0 (SOC 2010): Ref. 1 = BLS ISCO-SOC; Ref. 2 = ESCO-O*NET</i>			
19-3022 Survey Researchers	4227 Survey and Market Research Interviewers	2120 Mathematicians, Actuaries and Statisticians	—
11-3131 Training and Development Managers	2424 Training and Staff Development Professionals	1212 Human Resource Managers	2424 Training and Staff Development Professionals
41-3041 Travel Agents	4221 Travel Consultants and Clerks	3339 Business Services Agents Not Elsewhere Classified	4221 Travel Consultants and Clerks
13-1151 Training and Development Specialists	2424 Training and Staff Development Professionals	2356 Information Technology Trainers	2424 Training and Staff Development Professionals
<i>O*NET 29.2 (SOC 2018): Ref. 1 = ONET-SOC-ESCO; Ref. 2 = ESCO-ONET-SOC</i>			
19-3022 Survey Researchers	4227 Survey and Market Research Interviewers	2120 Mathematicians, Actuaries and Statisticians	2120 Mathematicians, Actuaries and Statisticians
11-3131 Training and Development Managers	2424 Training and Staff Development Professionals	1212 Human Resource Managers	2359 Teaching Professionals Not Elsewhere Classified
43-9081 Proofreaders and Copy Markers	4413 Coding, Proofreading and Related Clerks	2642 Journalists	1349 Professional Services Managers Not Elsewhere Classified
17-3031 Surveying and Mapping Technicians	2165 Cartographers and Surveyors	3112 Civil Engineering Technicians	2165 Cartographers and Surveyors

D Author annotation of O*NET 29.2 task assignments

To complement the automated chain-crosswalk validation, 108 O*NET 29.2 tasks were selected and annotated by the author. Tasks were drawn from 36 SOC occupations covering all nine ISCO major groups, with occupations chosen in order of combined US and Nordic employment within each group and tasks selected by O*NET importance score (top three per occupation). For each task the author assigned the most appropriate ISCO-08 four-digit unit group from a shortlist of plausible candidates, without reference to the pipeline’s predictions.

The evaluation below serves as a post-hoc consistency check on the unsupervised config selection performed in Appendix A. The selected configuration ($w_{\text{soc}} = 0.375$, $w_{\text{dwa}} = 0.266$, $w_{\text{isco}} = 0.375$, $w_{\text{isco-task}} = 0.734$, $w_{\text{occ}} = 0.156$, max links= 1) achieves 36.1% exact four-digit agreement with the author annotations, rising to 57.4% at the two-digit sub-major level and 72.2% at the one-digit major-group level, at a mean cosine similarity of 0.739. On the automated chain-crosswalk benchmark (lenient union A4), the selected configuration achieves 68.0% exact agreement.

These results confirm that the semantic similarity approach produces assignments broadly consistent with expert judgement at the occupation-group level. The author annotation reflects a single annotator’s judgements and should be interpreted as an approximate consistency check rather than a definitive gold standard; the automated chain-crosswalk benchmark, which is based on institutional SOC–ISCO concordances, is the primary validation criterion.

E Coverage and example matches

The pipeline was applied to all 63 O*NET releases from v4.0 (2001) through v30.3 (2025). For releases prior to v20.1, O*NET distributed data in text format only; Excel format is available from v20.1 onward. Occupation titles are merged from `Occupation Data.txt` for all pre-v20.1 releases so that occupational context is preserved in the query embedding across the full version range. All parameters and threshold values are identical across versions; only the input data (task list and SOC taxonomy generation) differs.

For O*NET 29.2 (the release used for parameter selection), the mapping assigns tasks to 435 of the 436 ISCO-08 unit groups. The only group without assigned tasks is ISCO 7516 (Tobacco Preparers and Tobacco Products Makers), which has no functional equivalent in O*NET task descriptions.

Coverage is broad across major occupational groups, including professional, technical, clerical, service, and manual occupations. Differences in task counts across ISCO-08 groups reflect differences in the level of detail in O*NET task descriptions and structural differences between U.S. and international occupational classifications.

Figure E1 shows how coverage and mean similarity evolve across the three pipeline stages across all O*NET releases. The retrieval stage (S1) achieves broad coverage at the cost of many candidates per task; subsequent filtering stages progressively tighten precision while maintaining near-complete coverage. Differences in task counts across ISCO-08 groups reflect structural differences between the SOC and ISCO-08 taxonomies rather than assignment errors; groups with the highest counts draw over 75% of their tasks from a single SOC major group (see Table E3).

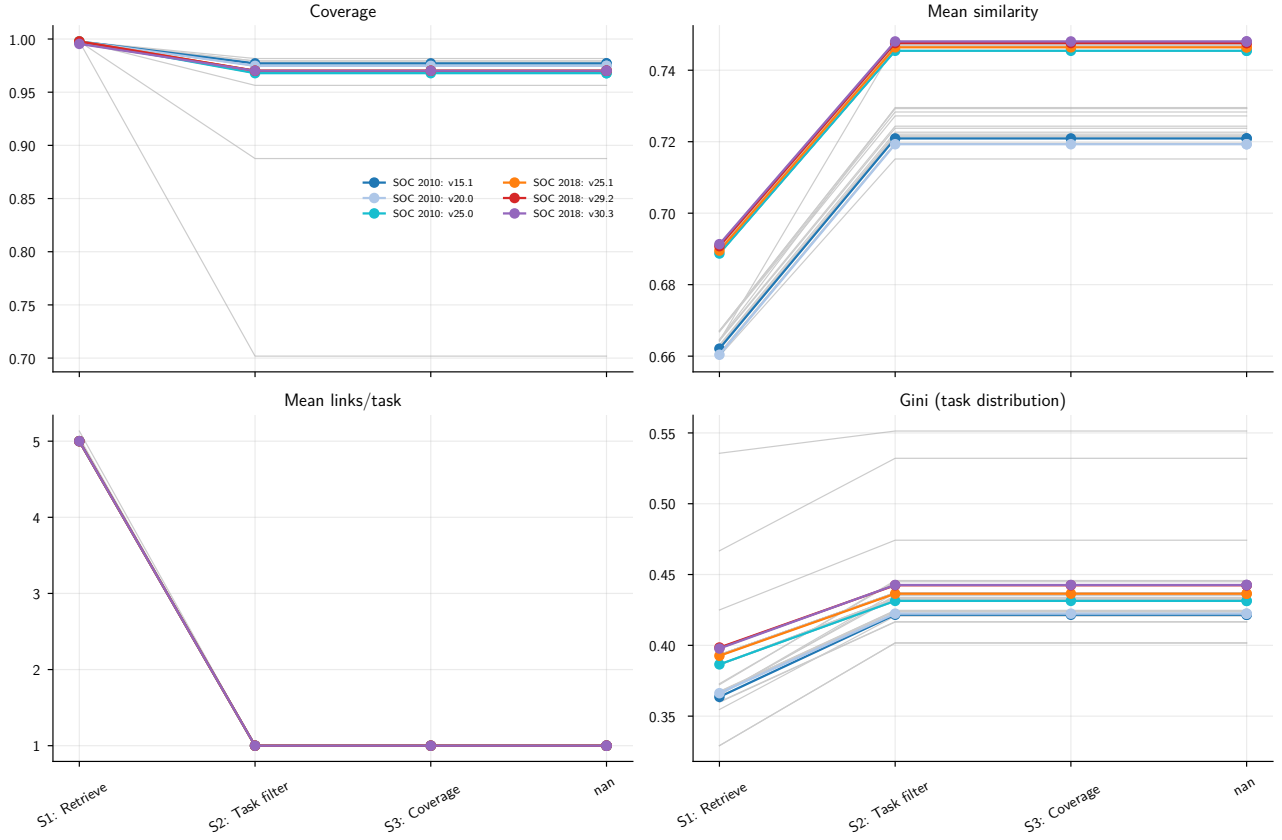


Figure E1: Pipeline metrics at each processing stage (S1–S3) across all O*NET releases (v4.0–v30.3).

Table E1 reports the corresponding numeric values for selected releases spanning the full version range.

Table E1: Pipeline metrics by stage and dataset.

Dataset	Stage	Cov.	Sim.	Links	Gini
15.100000	S1: Retrieve	0.998	0.662	5.000	0.364
15.100000	S2: Task filter	0.977	0.721	1.000	0.422
15.100000	S3: Coverage	0.977	0.721	1.000	0.422
15.100000	NaN	0.977	0.721	1.000	0.422
15.100000	NaN	0.977	0.721	1.000	0.422
20.000000	S1: Retrieve	0.998	0.660	5.000	0.366
20.000000	S2: Task filter	0.975	0.719	1.000	0.422
20.000000	S3: Coverage	0.975	0.719	1.000	0.422
20.000000	NaN	0.975	0.719	1.000	0.422
20.000000	NaN	0.975	0.719	1.000	0.422
25.000000	S1: Retrieve	0.998	0.689	5.000	0.387
25.000000	S2: Task filter	0.968	0.745	1.000	0.431
25.000000	S3: Coverage	0.968	0.745	1.000	0.431
25.000000	NaN	0.968	0.745	1.000	0.431
25.000000	NaN	0.968	0.745	1.000	0.431
25.100000	S1: Retrieve	0.998	0.690	5.000	0.393
25.100000	S2: Task filter	0.970	0.746	1.000	0.436
25.100000	S3: Coverage	0.970	0.746	1.000	0.436
25.100000	NaN	0.970	0.746	1.000	0.436
25.100000	NaN	0.970	0.746	1.000	0.436
29.200000	S1: Retrieve	0.998	0.691	5.000	0.398
29.200000	S2: Task filter	0.970	0.748	1.000	0.442

Continued on next page

Table E1: Pipeline metrics by stage and dataset.

Dataset	Stage	Cov.	Sim.	Links	Gini
29.200000	S3: Coverage	0.970	0.748	1.000	0.442
29.200000	NaN	0.970	0.748	1.000	0.442
29.200000	NaN	0.970	0.748	1.000	0.442
30.300000	S1: Retrieve	0.995	0.691	5.000	0.398
30.300000	S2: Task filter	0.970	0.748	1.000	0.443
30.300000	S3: Coverage	0.970	0.748	1.000	0.443
30.300000	NaN	0.970	0.748	1.000	0.443
30.300000	NaN	0.970	0.748	1.000	0.443

Figure E2 compares final-stage (S3) summary metrics across all O*NET releases.

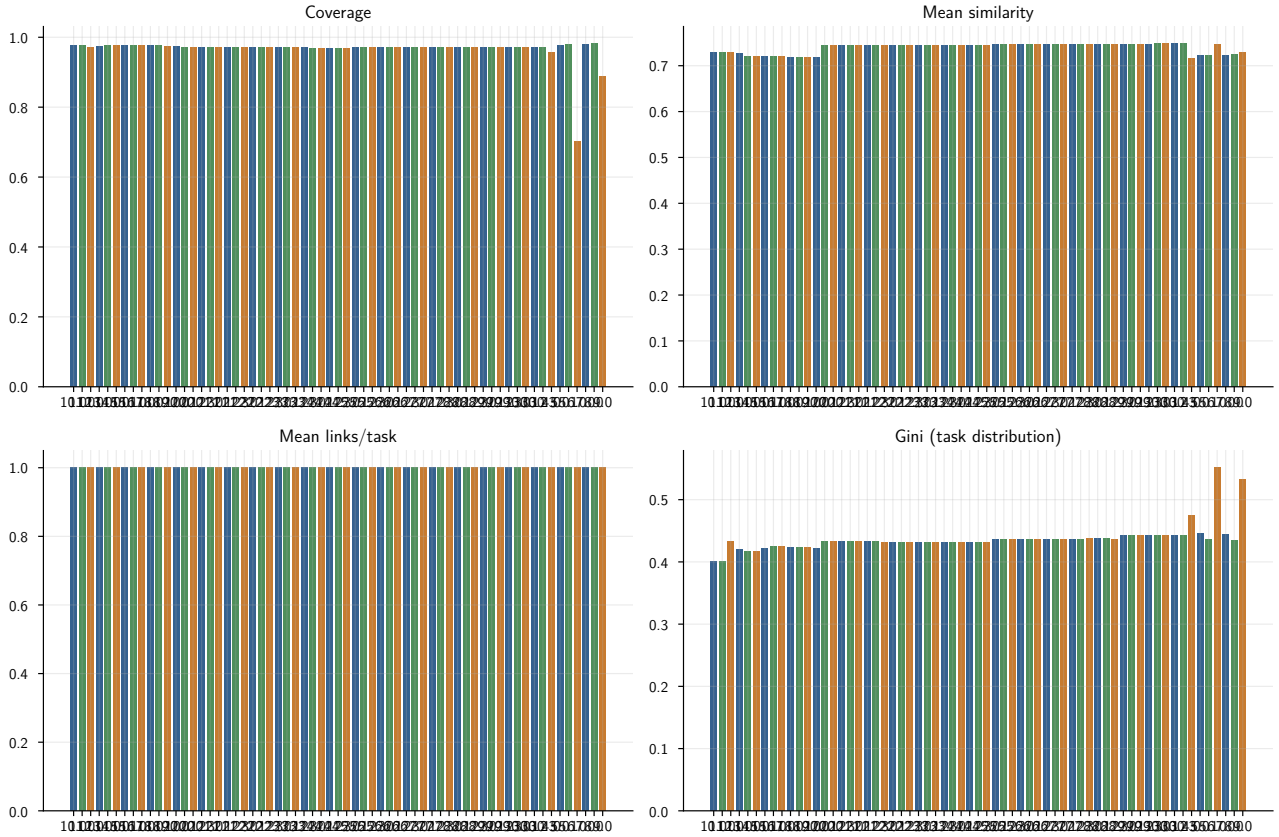


Figure E2: Final-stage (S3) metrics across all O*NET releases (v4.0–v30.3).

Table E2 summarises the S3 metrics numerically.

Table E2: Final-stage (S3) summary metrics across all O*NET releases (v4.0–v30.3).

Dataset	Cov.	Sim.	Links	Gini
15.100000	0.977	0.721	1.000	0.422
20.000000	0.975	0.719	1.000	0.422
25.000000	0.968	0.745	1.000	0.431
25.100000	0.970	0.746	1.000	0.436
29.200000	0.970	0.748	1.000	0.442
30.300000	0.970	0.748	1.000	0.443

Table E3 lists the ISCO-08 unit groups that attract the largest numbers of task assignments in the final output across all O*NET releases. These groups are not catch-all categories; they are structurally

dense nodes where the SOC taxonomy distinguishes many occupations that ISCO-08 unifies into a single unit group.

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
4.000000	7223	Metal Working Machine Tool Sette	342
4.000000	7233	Agricultural and Industrial Mach	157
4.000000	3341	Office Supervisors	121
4.000000	7222	Toolmakers and Related Workers	111
4.000000	7412	Electrical Mechanics and Fitters	104
4.000000	7231	Motor Vehicle Mechanics and Repa	93
4.000000	7421	Electronics Mechanics and Servic	85
4.000000	8132	Photographic Products Machine Op	84
5.000000	7223	Metal Working Machine Tool Sette	342
5.000000	7233	Agricultural and Industrial Mach	181
5.000000	3341	Office Supervisors	145
5.000000	5411	Fire Fighters	134
5.000000	7231	Motor Vehicle Mechanics and Repa	114
5.000000	3211	Medical Imaging and Therapeutic	113
5.000000	7421	Electronics Mechanics and Servic	111
5.000000	7222	Toolmakers and Related Workers	111
5.100000	7223	Metal Working Machine Tool Sette	341
5.100000	7233	Agricultural and Industrial Mach	181
5.100000	3341	Office Supervisors	145
5.100000	5411	Fire Fighters	134
5.100000	7231	Motor Vehicle Mechanics and Repa	115
5.100000	3211	Medical Imaging and Therapeutic	113
5.100000	7222	Toolmakers and Related Workers	111
5.100000	7421	Electronics Mechanics and Servic	111
6.000000	3211	Medical Imaging and Therapeutic	59
6.000000	7233	Agricultural and Industrial Mach	56
6.000000	7231	Motor Vehicle Mechanics and Repa	53
6.000000	3355	Police Inspectors and Detectives	48
6.000000	4411	Library Clerks	48
6.000000	5329	Personal Care Workers in Health	47
6.000000	3323	Buyers	46
6.000000	5223	Shop Sales Assistants	44
7.000000	7223	Metal Working Machine Tool Sette	395
7.000000	7233	Agricultural and Industrial Mach	202
7.000000	3341	Office Supervisors	187
7.000000	7421	Electronics Mechanics and Servic	150
7.000000	8132	Photographic Products Machine Op	139
7.000000	5411	Fire Fighters	138
7.000000	7412	Electrical Mechanics and Fitters	130
7.000000	7222	Toolmakers and Related Workers	129
8.000000	7223	Metal Working Machine Tool Sette	382
8.000000	7233	Agricultural and Industrial Mach	220
8.000000	3341	Office Supervisors	190
8.000000	7421	Electronics Mechanics and Servic	153
8.000000	5411	Fire Fighters	144
8.000000	8132	Photographic Products Machine Op	139
8.000000	7222	Toolmakers and Related Workers	129
8.000000	7412	Electrical Mechanics and Fitters	128
9.000000	2310	University and Higher Education	267
9.000000	1345	Education Managers	261
9.000000	2351	Education Methods Specialists	156

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
9.000000	5411	Fire Fighters	113
9.000000	3211	Medical Imaging and Therapeutic	110
9.000000	2359	Teaching Professionals Not Elsew	108
9.000000	2352	Special Needs Teachers	104
9.000000	7231	Motor Vehicle Mechanics and Repa	90
10.000000	7223	Metal Working Machine Tool Sette	215
10.000000	7233	Agricultural and Industrial Mach	202
10.000000	5411	Fire Fighters	143
10.000000	3341	Office Supervisors	143
10.000000	7421	Electronics Mechanics and Servic	121
10.000000	7231	Motor Vehicle Mechanics and Repa	119
10.000000	3211	Medical Imaging and Therapeutic	112
10.000000	9329	Manufacturing Labourers Not Else	105
11.000000	7223	Metal Working Machine Tool Sette	215
11.000000	7233	Agricultural and Industrial Mach	202
11.000000	5411	Fire Fighters	143
11.000000	3341	Office Supervisors	143
11.000000	7421	Electronics Mechanics and Servic	121
11.000000	7231	Motor Vehicle Mechanics and Repa	119
11.000000	3211	Medical Imaging and Therapeutic	112
11.000000	9329	Manufacturing Labourers Not Else	105
12.000000	2310	University and Higher Education	279
12.000000	1345	Education Managers	266
12.000000	7223	Metal Working Machine Tool Sette	210
12.000000	7233	Agricultural and Industrial Mach	205
12.000000	2351	Education Methods Specialists	158
12.000000	5411	Fire Fighters	144
12.000000	3341	Office Supervisors	142
12.000000	7421	Electronics Mechanics and Servic	122
13.000000	2310	University and Higher Education	271
13.000000	1345	Education Managers	268
13.000000	7223	Metal Working Machine Tool Sette	215
13.000000	7233	Agricultural and Industrial Mach	205
13.000000	2351	Education Methods Specialists	165
13.000000	5411	Fire Fighters	143
13.000000	3341	Office Supervisors	142
13.000000	7421	Electronics Mechanics and Servic	123
14.000000	2310	University and Higher Education	275
14.000000	1345	Education Managers	273
14.000000	7223	Metal Working Machine Tool Sette	215
14.000000	7233	Agricultural and Industrial Mach	209
14.000000	2351	Education Methods Specialists	171
14.000000	3211	Medical Imaging and Therapeutic	164
14.000000	3131	Power Production Plant Operators	162
14.000000	2212	Specialist Medical Practitioners	161
15.000000	1345	Education Managers	280
15.000000	2310	University and Higher Education	275
15.000000	7223	Metal Working Machine Tool Sette	216
15.000000	7233	Agricultural and Industrial Mach	211
15.000000	2351	Education Methods Specialists	171
15.000000	3131	Power Production Plant Operators	164
15.000000	3211	Medical Imaging and Therapeutic	164
15.000000	2212	Specialist Medical Practitioners	161
15.100000	1345	Education Managers	283

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
15.100000	2310	University and Higher Education	275
15.100000	7223	Metal Working Machine Tool Sette	221
15.100000	7233	Agricultural and Industrial Mach	204
15.100000	2351	Education Methods Specialists	176
15.100000	3211	Medical Imaging and Therapeutic	176
15.100000	5329	Personal Care Workers in Health	170
15.100000	3131	Power Production Plant Operators	168
16.000000	1345	Education Managers	285
16.000000	2310	University and Higher Education	281
16.000000	7223	Metal Working Machine Tool Sette	222
16.000000	7233	Agricultural and Industrial Mach	200
16.000000	3131	Power Production Plant Operators	180
16.000000	2351	Education Methods Specialists	178
16.000000	3211	Medical Imaging and Therapeutic	177
16.000000	5329	Personal Care Workers in Health	170
17.000000	1345	Education Managers	285
17.000000	2310	University and Higher Education	281
17.000000	7223	Metal Working Machine Tool Sette	223
17.000000	7233	Agricultural and Industrial Mach	197
17.000000	3131	Power Production Plant Operators	180
17.000000	2351	Education Methods Specialists	178
17.000000	3211	Medical Imaging and Therapeutic	177
17.000000	5329	Personal Care Workers in Health	170
18.000000	1345	Education Managers	285
18.000000	2310	University and Higher Education	281
18.000000	7223	Metal Working Machine Tool Sette	223
18.000000	7233	Agricultural and Industrial Mach	198
18.000000	3131	Power Production Plant Operators	180
18.000000	3211	Medical Imaging and Therapeutic	180
18.000000	2351	Education Methods Specialists	178
18.000000	5329	Personal Care Workers in Health	170
18.100000	1345	Education Managers	285
18.100000	2310	University and Higher Education	281
18.100000	7223	Metal Working Machine Tool Sette	223
18.100000	7233	Agricultural and Industrial Mach	198
18.100000	3131	Power Production Plant Operators	180
18.100000	3211	Medical Imaging and Therapeutic	180
18.100000	2351	Education Methods Specialists	178
18.100000	5329	Personal Care Workers in Health	170
19.000000	1345	Education Managers	285
19.000000	2310	University and Higher Education	281
19.000000	7223	Metal Working Machine Tool Sette	223
19.000000	7233	Agricultural and Industrial Mach	200
19.000000	3131	Power Production Plant Operators	181
19.000000	3211	Medical Imaging and Therapeutic	180
19.000000	2351	Education Methods Specialists	178
19.000000	5329	Personal Care Workers in Health	171
20.000000	1345	Education Managers	287
20.000000	2310	University and Higher Education	281
20.000000	7223	Metal Working Machine Tool Sette	223
20.000000	7233	Agricultural and Industrial Mach	200
20.000000	3131	Power Production Plant Operators	181
20.000000	3211	Medical Imaging and Therapeutic	180
20.000000	2351	Education Methods Specialists	178

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
20.000000	5329	Personal Care Workers in Health	171
20.100000	2310	University and Higher Education	284
20.100000	1345	Education Managers	270
20.100000	7223	Metal Working Machine Tool Sette	206
20.100000	7233	Agricultural and Industrial Mach	206
20.100000	2351	Education Methods Specialists	193
20.100000	5329	Personal Care Workers in Health	189
20.100000	3211	Medical Imaging and Therapeutic	184
20.100000	2359	Teaching Professionals Not Elsew	181
20.200000	2310	University and Higher Education	284
20.200000	1345	Education Managers	270
20.200000	7223	Metal Working Machine Tool Sette	206
20.200000	7233	Agricultural and Industrial Mach	206
20.200000	2351	Education Methods Specialists	193
20.200000	5329	Personal Care Workers in Health	189
20.200000	3211	Medical Imaging and Therapeutic	184
20.200000	2359	Teaching Professionals Not Elsew	181
20.300000	2310	University and Higher Education	284
20.300000	1345	Education Managers	270
20.300000	7223	Metal Working Machine Tool Sette	206
20.300000	7233	Agricultural and Industrial Mach	206
20.300000	2351	Education Methods Specialists	193
20.300000	5329	Personal Care Workers in Health	189
20.300000	3211	Medical Imaging and Therapeutic	184
20.300000	2359	Teaching Professionals Not Elsew	181
21.000000	2310	University and Higher Education	285
21.000000	1345	Education Managers	270
21.000000	7233	Agricultural and Industrial Mach	208
21.000000	7223	Metal Working Machine Tool Sette	206
21.000000	2351	Education Methods Specialists	190
21.000000	5329	Personal Care Workers in Health	189
21.000000	3211	Medical Imaging and Therapeutic	184
21.000000	2359	Teaching Professionals Not Elsew	181
21.100000	2310	University and Higher Education	285
21.100000	1345	Education Managers	270
21.100000	7233	Agricultural and Industrial Mach	208
21.100000	7223	Metal Working Machine Tool Sette	206
21.100000	2351	Education Methods Specialists	190
21.100000	5329	Personal Care Workers in Health	189
21.100000	3211	Medical Imaging and Therapeutic	184
21.100000	2359	Teaching Professionals Not Elsew	181
21.200000	2310	University and Higher Education	285
21.200000	1345	Education Managers	270
21.200000	7233	Agricultural and Industrial Mach	208
21.200000	7223	Metal Working Machine Tool Sette	206
21.200000	2351	Education Methods Specialists	190
21.200000	5329	Personal Care Workers in Health	189
21.200000	3211	Medical Imaging and Therapeutic	184
21.200000	2359	Teaching Professionals Not Elsew	181
21.300000	2310	University and Higher Education	285
21.300000	1345	Education Managers	270
21.300000	7233	Agricultural and Industrial Mach	208
21.300000	7223	Metal Working Machine Tool Sette	206
21.300000	2351	Education Methods Specialists	190

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
21.300000	5329	Personal Care Workers in Health	189
21.300000	3211	Medical Imaging and Therapeutic	184
21.300000	2359	Teaching Professionals Not Elsew	181
22.000000	2310	University and Higher Education	285
22.000000	1345	Education Managers	270
22.000000	7233	Agricultural and Industrial Mach	207
22.000000	7223	Metal Working Machine Tool Sette	206
22.000000	2351	Education Methods Specialists	190
22.000000	5329	Personal Care Workers in Health	189
22.000000	3211	Medical Imaging and Therapeutic	184
22.000000	2359	Teaching Professionals Not Elsew	181
22.100000	2310	University and Higher Education	285
22.100000	1345	Education Managers	270
22.100000	7233	Agricultural and Industrial Mach	207
22.100000	7223	Metal Working Machine Tool Sette	206
22.100000	2351	Education Methods Specialists	190
22.100000	5329	Personal Care Workers in Health	189
22.100000	3211	Medical Imaging and Therapeutic	184
22.100000	2359	Teaching Professionals Not Elsew	181
22.200000	2310	University and Higher Education	285
22.200000	1345	Education Managers	270
22.200000	7233	Agricultural and Industrial Mach	207
22.200000	7223	Metal Working Machine Tool Sette	206
22.200000	2351	Education Methods Specialists	190
22.200000	5329	Personal Care Workers in Health	189
22.200000	3211	Medical Imaging and Therapeutic	184
22.200000	2359	Teaching Professionals Not Elsew	181
22.300000	2310	University and Higher Education	285
22.300000	1345	Education Managers	270
22.300000	7233	Agricultural and Industrial Mach	207
22.300000	7223	Metal Working Machine Tool Sette	206
22.300000	2351	Education Methods Specialists	190
22.300000	5329	Personal Care Workers in Health	189
22.300000	3211	Medical Imaging and Therapeutic	184
22.300000	2359	Teaching Professionals Not Elsew	181
23.000000	2310	University and Higher Education	286
23.000000	1345	Education Managers	269
23.000000	7233	Agricultural and Industrial Mach	207
23.000000	7223	Metal Working Machine Tool Sette	206
23.000000	2351	Education Methods Specialists	191
23.000000	5329	Personal Care Workers in Health	188
23.000000	3211	Medical Imaging and Therapeutic	185
23.000000	2359	Teaching Professionals Not Elsew	181
23.100000	2310	University and Higher Education	286
23.100000	1345	Education Managers	269
23.100000	7233	Agricultural and Industrial Mach	207
23.100000	7223	Metal Working Machine Tool Sette	206
23.100000	2351	Education Methods Specialists	191
23.100000	5329	Personal Care Workers in Health	188
23.100000	3211	Medical Imaging and Therapeutic	185
23.100000	2359	Teaching Professionals Not Elsew	181
23.200000	2310	University and Higher Education	286
23.200000	1345	Education Managers	269
23.200000	7233	Agricultural and Industrial Mach	207

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
23.200000	7223	Metal Working Machine Tool Sette	206
23.200000	2351	Education Methods Specialists	191
23.200000	5329	Personal Care Workers in Health	188
23.200000	3211	Medical Imaging and Therapeutic	185
23.200000	2359	Teaching Professionals Not Elsew	181
23.300000	2310	University and Higher Education	286
23.300000	1345	Education Managers	269
23.300000	7233	Agricultural and Industrial Mach	207
23.300000	7223	Metal Working Machine Tool Sette	206
23.300000	2351	Education Methods Specialists	191
23.300000	5329	Personal Care Workers in Health	188
23.300000	3211	Medical Imaging and Therapeutic	185
23.300000	2359	Teaching Professionals Not Elsew	181
24.000000	2310	University and Higher Education	288
24.000000	1345	Education Managers	271
24.000000	7233	Agricultural and Industrial Mach	207
24.000000	7223	Metal Working Machine Tool Sette	206
24.000000	2351	Education Methods Specialists	195
24.000000	5329	Personal Care Workers in Health	187
24.000000	3211	Medical Imaging and Therapeutic	186
24.000000	2359	Teaching Professionals Not Elsew	186
24.100000	2310	University and Higher Education	288
24.100000	1345	Education Managers	271
24.100000	7233	Agricultural and Industrial Mach	207
24.100000	7223	Metal Working Machine Tool Sette	206
24.100000	2351	Education Methods Specialists	195
24.100000	5329	Personal Care Workers in Health	187
24.100000	3211	Medical Imaging and Therapeutic	186
24.100000	2359	Teaching Professionals Not Elsew	186
24.200000	2310	University and Higher Education	288
24.200000	1345	Education Managers	271
24.200000	7233	Agricultural and Industrial Mach	207
24.200000	7223	Metal Working Machine Tool Sette	206
24.200000	2351	Education Methods Specialists	195
24.200000	5329	Personal Care Workers in Health	187
24.200000	3211	Medical Imaging and Therapeutic	186
24.200000	2359	Teaching Professionals Not Elsew	186
24.300000	2310	University and Higher Education	288
24.300000	1345	Education Managers	271
24.300000	7233	Agricultural and Industrial Mach	207
24.300000	7223	Metal Working Machine Tool Sette	206
24.300000	2351	Education Methods Specialists	195
24.300000	5329	Personal Care Workers in Health	187
24.300000	3211	Medical Imaging and Therapeutic	186
24.300000	2359	Teaching Professionals Not Elsew	186
25.000000	2310	University and Higher Education	289
25.000000	1345	Education Managers	271
25.000000	7233	Agricultural and Industrial Mach	207
25.000000	7223	Metal Working Machine Tool Sette	206
25.000000	2351	Education Methods Specialists	195
25.000000	5329	Personal Care Workers in Health	188
25.000000	3211	Medical Imaging and Therapeutic	186
25.000000	2359	Teaching Professionals Not Elsew	186
25.100000	2310	University and Higher Education	290

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
25.100000	1345	Education Managers	256
25.100000	2212	Specialist Medical Practitioners	202
25.100000	2351	Education Methods Specialists	202
25.100000	7233	Agricultural and Industrial Mach	201
25.100000	7223	Metal Working Machine Tool Sette	193
25.100000	2359	Teaching Professionals Not Elsew	189
25.100000	3341	Office Supervisors	186
25.200000	2310	University and Higher Education	290
25.200000	1345	Education Managers	256
25.200000	2212	Specialist Medical Practitioners	202
25.200000	2351	Education Methods Specialists	202
25.200000	7233	Agricultural and Industrial Mach	201
25.200000	7223	Metal Working Machine Tool Sette	193
25.200000	2359	Teaching Professionals Not Elsew	189
25.200000	3341	Office Supervisors	186
25.300000	2310	University and Higher Education	290
25.300000	1345	Education Managers	256
25.300000	2212	Specialist Medical Practitioners	202
25.300000	2351	Education Methods Specialists	202
25.300000	7233	Agricultural and Industrial Mach	201
25.300000	7223	Metal Working Machine Tool Sette	193
25.300000	2359	Teaching Professionals Not Elsew	189
25.300000	3341	Office Supervisors	186
26.000000	2310	University and Higher Education	289
26.000000	1345	Education Managers	257
26.000000	2351	Education Methods Specialists	202
26.000000	2212	Specialist Medical Practitioners	202
26.000000	7233	Agricultural and Industrial Mach	199
26.000000	7223	Metal Working Machine Tool Sette	194
26.000000	3341	Office Supervisors	189
26.000000	2359	Teaching Professionals Not Elsew	188
26.100000	2310	University and Higher Education	289
26.100000	1345	Education Managers	257
26.100000	2351	Education Methods Specialists	202
26.100000	2212	Specialist Medical Practitioners	202
26.100000	7233	Agricultural and Industrial Mach	199
26.100000	7223	Metal Working Machine Tool Sette	194
26.100000	3341	Office Supervisors	189
26.100000	2359	Teaching Professionals Not Elsew	188
26.200000	2310	University and Higher Education	289
26.200000	1345	Education Managers	257
26.200000	2351	Education Methods Specialists	202
26.200000	2212	Specialist Medical Practitioners	202
26.200000	7233	Agricultural and Industrial Mach	199
26.200000	7223	Metal Working Machine Tool Sette	194
26.200000	3341	Office Supervisors	189
26.200000	2359	Teaching Professionals Not Elsew	188
26.300000	2310	University and Higher Education	289
26.300000	1345	Education Managers	257
26.300000	2351	Education Methods Specialists	202
26.300000	2212	Specialist Medical Practitioners	202
26.300000	7233	Agricultural and Industrial Mach	199
26.300000	7223	Metal Working Machine Tool Sette	194
26.300000	3341	Office Supervisors	189

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
26.300000	2359	Teaching Professionals Not Elsew	188
27.000000	2310	University and Higher Education	290
27.000000	1345	Education Managers	257
27.000000	2212	Specialist Medical Practitioners	202
27.000000	2351	Education Methods Specialists	200
27.000000	7233	Agricultural and Industrial Mach	199
27.000000	7223	Metal Working Machine Tool Sette	194
27.000000	3341	Office Supervisors	189
27.000000	2359	Teaching Professionals Not Elsew	188
27.100000	2310	University and Higher Education	290
27.100000	1345	Education Managers	257
27.100000	2212	Specialist Medical Practitioners	202
27.100000	2351	Education Methods Specialists	200
27.100000	7233	Agricultural and Industrial Mach	199
27.100000	7223	Metal Working Machine Tool Sette	194
27.100000	3341	Office Supervisors	189
27.100000	2359	Teaching Professionals Not Elsew	188
27.200000	2310	University and Higher Education	290
27.200000	1345	Education Managers	257
27.200000	2212	Specialist Medical Practitioners	202
27.200000	2351	Education Methods Specialists	200
27.200000	7233	Agricultural and Industrial Mach	199
27.200000	7223	Metal Working Machine Tool Sette	194
27.200000	3341	Office Supervisors	189
27.200000	2359	Teaching Professionals Not Elsew	188
27.300000	2310	University and Higher Education	290
27.300000	1345	Education Managers	257
27.300000	2212	Specialist Medical Practitioners	202
27.300000	2351	Education Methods Specialists	200
27.300000	7233	Agricultural and Industrial Mach	199
27.300000	7223	Metal Working Machine Tool Sette	194
27.300000	3341	Office Supervisors	189
27.300000	2359	Teaching Professionals Not Elsew	188
28.000000	2310	University and Higher Education	290
28.000000	1345	Education Managers	257
28.000000	2212	Specialist Medical Practitioners	202
28.000000	2351	Education Methods Specialists	200
28.000000	7233	Agricultural and Industrial Mach	199
28.000000	7223	Metal Working Machine Tool Sette	194
28.000000	3341	Office Supervisors	188
28.000000	2359	Teaching Professionals Not Elsew	188
28.100000	2310	University and Higher Education	290
28.100000	1345	Education Managers	257
28.100000	2212	Specialist Medical Practitioners	202
28.100000	2351	Education Methods Specialists	200
28.100000	7233	Agricultural and Industrial Mach	199
28.100000	7223	Metal Working Machine Tool Sette	194
28.100000	3341	Office Supervisors	188
28.100000	2359	Teaching Professionals Not Elsew	188
28.200000	2310	University and Higher Education	290
28.200000	1345	Education Managers	257
28.200000	2212	Specialist Medical Practitioners	202
28.200000	2351	Education Methods Specialists	200
28.200000	7233	Agricultural and Industrial Mach	199

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
28.200000	7223	Metal Working Machine Tool Sette	194
28.200000	3341	Office Supervisors	188
28.200000	2359	Teaching Professionals Not Elsew	188
28.300000	2310	University and Higher Education	290
28.300000	1345	Education Managers	257
28.300000	2212	Specialist Medical Practitioners	202
28.300000	2351	Education Methods Specialists	200
28.300000	7233	Agricultural and Industrial Mach	199
28.300000	7223	Metal Working Machine Tool Sette	194
28.300000	3341	Office Supervisors	188
28.300000	2359	Teaching Professionals Not Elsew	188
29.000000	2310	University and Higher Education	290
29.000000	1345	Education Managers	255
29.000000	2212	Specialist Medical Practitioners	200
29.000000	2351	Education Methods Specialists	198
29.000000	7233	Agricultural and Industrial Mach	195
29.000000	7223	Metal Working Machine Tool Sette	191
29.000000	2359	Teaching Professionals Not Elsew	188
29.000000	3341	Office Supervisors	186
29.100000	2310	University and Higher Education	290
29.100000	1345	Education Managers	255
29.100000	2212	Specialist Medical Practitioners	200
29.100000	2351	Education Methods Specialists	198
29.100000	7233	Agricultural and Industrial Mach	195
29.100000	7223	Metal Working Machine Tool Sette	191
29.100000	2359	Teaching Professionals Not Elsew	188
29.100000	3341	Office Supervisors	186
29.200000	2310	University and Higher Education	290
29.200000	1345	Education Managers	255
29.200000	2212	Specialist Medical Practitioners	200
29.200000	2351	Education Methods Specialists	198
29.200000	7233	Agricultural and Industrial Mach	195
29.200000	7223	Metal Working Machine Tool Sette	191
29.200000	2359	Teaching Professionals Not Elsew	188
29.200000	3341	Office Supervisors	186
29.300000	2310	University and Higher Education	290
29.300000	1345	Education Managers	255
29.300000	2212	Specialist Medical Practitioners	200
29.300000	2351	Education Methods Specialists	198
29.300000	7233	Agricultural and Industrial Mach	195
29.300000	7223	Metal Working Machine Tool Sette	191
29.300000	2359	Teaching Professionals Not Elsew	188
29.300000	3341	Office Supervisors	186
30.000000	2310	University and Higher Education	290
30.000000	1345	Education Managers	257
30.000000	2212	Specialist Medical Practitioners	201
30.000000	2351	Education Methods Specialists	198
30.000000	7233	Agricultural and Industrial Mach	195
30.000000	7223	Metal Working Machine Tool Sette	188
30.000000	2359	Teaching Professionals Not Elsew	188
30.000000	3341	Office Supervisors	186
30.100000	2310	University and Higher Education	290
30.100000	1345	Education Managers	257
30.100000	2212	Specialist Medical Practitioners	201

Continued on next page

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, shown for all O*NET releases.

Dataset	ISCO	Occupation label	Tasks
30.100000	2351	Education Methods Specialists	198
30.100000	7233	Agricultural and Industrial Mach	195
30.100000	7223	Metal Working Machine Tool Sette	188
30.100000	2359	Teaching Professionals Not Elsew	188
30.100000	3341	Office Supervisors	186
30.200000	2310	University and Higher Education	290
30.200000	1345	Education Managers	257
30.200000	2212	Specialist Medical Practitioners	201
30.200000	2351	Education Methods Specialists	198
30.200000	7233	Agricultural and Industrial Mach	195
30.200000	7223	Metal Working Machine Tool Sette	188
30.200000	2359	Teaching Professionals Not Elsew	188
30.200000	3341	Office Supervisors	186
30.300000	2310	University and Higher Education	290
30.300000	1345	Education Managers	257
30.300000	2212	Specialist Medical Practitioners	201
30.300000	2351	Education Methods Specialists	198
30.300000	7233	Agricultural and Industrial Mach	195
30.300000	7223	Metal Working Machine Tool Sette	188
30.300000	2359	Teaching Professionals Not Elsew	188
30.300000	3341	Office Supervisors	186

Table E4 reports illustrative task assignments, one per ISCO major group, showing the full query-side inputs (task text, SOC occupation, DWA labels) alongside the pipeline assignment.

Table E4: Examples of task assignments (one per ISCO major group). O*NET task text, SOC occupation title, and the top two DWA activity labels constitute the query-side inputs; the ISCO-08 assignment is the pipeline output.

O*NET Task	SOC Occupation	DWA Labels (top 2)	ISCO-08 Assignment
Develop and implement organizational policies and procedures for the facility or medical unit	Medical and Health Services Managers	Develop operating strategies, plans, or procedures.; Develop organizational policies or programs	1342 Health Service Managers
Prepare, analyze, or verify annual reports, financial statements, and other records, using accepted accounting and statistical procedures to assess financial condition and facilitate financial planning	Accountants and Auditors	Prepare financial documents, reports, or budgets.; Verify accuracy of financial information	2411 Accountants
Obtain and verify evidence by interviewing and observing suspects and witnesses or by analyzing records	Detectives and Criminal Investigators	Examine records or other types of data to investigate criminal activities.; Interview people to gather information about criminal activities	3355 Police Inspectors and Detectives
Process and issue employee paychecks and statements of earnings and deductions	Payroll and Time-keeping Clerks	Execute sales or other financial transactions	4313 Payroll Clerks
Monitor conduct of prisoners in housing unit, or during work or recreational activities, according to established policies, regulations, and procedures, to prevent escape or violence	Correctional Officers and Jailers	Maintain surveillance of individuals or establishments	5413 Prison Guards
Feed and water animals, and clean and disinfect pens, cages, yards, and hutches	Animal Breeders	Care for animals.; Clean equipment or facilities	6129 Animal Producers Not Elsewhere Classified
Operate newly repaired machinery or equipment to verify the adequacy of repairs	Industrial Machinery Mechanics	Test mechanical equipment to ensure proper functioning	7233 Agricultural and Industrial Machinery Mechanics and Repairers
Observe and interpret readings on gauges, meters, and charts registering various aspects of boiler operation to ensure that boilers are operating properly	Stationary Engineers and Boiler Operators	Monitor equipment operation to ensure proper functioning	8182 Steam Engine and Boiler Operators
Sort materials, such as metals, glass, wood, paper	Recycling and Reclamation	Sort recyclable materials	9612 Refuse Sorters